THE UNIVERSITY OF AUCKLAND

SECOND SEMESTER, 2024
Campus: City

MATHEMATICS

Topology

(Time allowed: TWO hours)

NOTE: This exam consists of **four questions**. Question 1 carries 10 marks and all other questions 15. The marks are given at the start of each question. The total number of marks is **55**.

Answer **all** questions.

Show all working for questions 2-4.

The Appendix at the end of the exam has the definition of a Urysohn family, and one lemma.

1. (10 marks)

   State whether each of the following statements is true or false. Give a brief justification for each answer, one or two lines. Fully detailed proofs are not required and counterexamples can be stated without proof.

   (a) Let
   $$\mathcal{U} = \{[a, b] : a, b \in \mathbb{R}, a < b\}$$
   be the subbasis for a topology τ on $\mathbb{R}$. Then τ is the finest topology on $\mathbb{R}$.

   (b) If X is a topological space and A is any subset of X, then
   $$\text{int}\,(\text{cl}\,(\text{int}\,(A))) = \text{int}\,A.$$

   (c) If X and Y are homeomorphic spaces, then any embedding of X into Y is surjective.

   (d) If $\mathbb{Z}$ has the usual discrete topology and $X = \mathbb{Z}^{\mathbb{N}}$ has the product topology, then X is discrete.

   (e) If X is a Hausdorff space then any singleton set is closed.

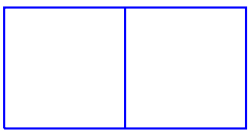
   (f) If X is a second countable space and $A \subseteq X$, then
   $$\bar{A} = \{x \in X : \text{there is a sequence } (x_n) \text{ in } A \text{ with } \lim_{n\to\infty} x_n = x\}.$$

   (g) A path-connected, connected space is locally connected.

   (h) If X and E are topological spaces and $p : E \to X$ is a covering projection, then for any path σ in X, there is a unique lifting of σ to a path in E.

   (i) The fundamental groups of the annulus and the cylinder are isomorphic.

   (j) The fundamental group of the structure shown in the diagram below is $\mathbb{Z} \oplus \mathbb{Z}$.



2. (15 marks)

   (a) Let X be an infinite set endowed with the co-finite topology. Show that X is:

   i. compact, and

   ii. not normal.

   (b) Let X and Y be topological spaces. Suppose that Y is Hausdorff and

   $$f, g : X \to Y$$

   are continuous functions. Show that the set

   $$\{x \in X : f(x) = g(x)\}$$

   is closed.

   (c) Prove that the product of two separable spaces is separable.

3. (15 marks)

   (a) Prove that the product of two connected topological spaces is connected.

   (b) For this question you may assume the result given in the Appendix.

   Let A and B be disjoint closed subsets of a space X. Prove that A and B are completely separated in X if, and only if, there is a Urysohn family $\{U_r : r \in D\}$ with $0, 1 \in D$, $A \subseteq U_0$ and $B \subseteq X \setminus U_1$.

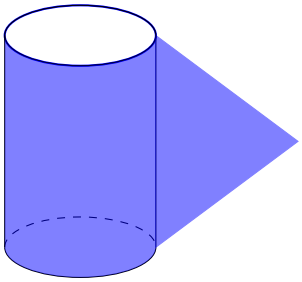
4. (15 marks)

   (a) Give two examples of covering spaces E and E' of the closed unit interval $[0, 1]$, together with their covering projections $p : E \to X$, and $p' : E' \to X$, such that

   i. E and E' are not homeomorphic, and

   ii. E and E' are connected.

   (b) Find the simplicial homology groups for the structure shown in the diagram below, a hollow cylinder with an attached triangle. Show all working.



Appendix

Definition Let X be a space. By a *Urysohn family* in X we mean a family $\{U_r : r \in D\}$ of open subsets of X satisfying:

(i) $\bar{D} = \mathbb{R}$;

(ii) $\bigcup_{r\in D} U_r = X$ and $\bigcap_{r\in D} U_r = \emptyset$;

(iii) if $r < s$ then $\overline{U_r} \subset U_s$.

Lemma Let $\mathcal{U} = \{U_r : r \in D\}$ be a Urysohn family in the space X and define a function $\phi_{\mathcal{U}} : X \to \mathbb{R}$ by

$$\phi_{\mathcal{U}}(x) = \inf\{r \in D : x \in U_r\}.$$

Then $\phi_{\mathcal{U}}$ is continuous.